

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	01 JULY 2026
Team ID	
Project Name	Credit card approval prediction
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Brainstorming & Ideation

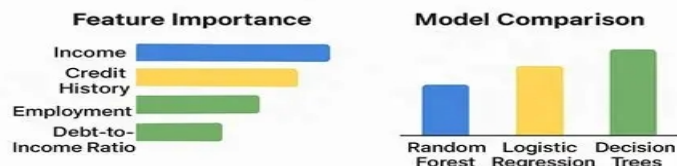
Matter:

- Problem Statement: Predict whether a credit card application should be approved based on applicant data.
- Ideas: Logistic Regression baseline, Random Forest, XGBoost, Flask API, Web frontend, fairness metrics.
- Grouping: Modeling, Deployment, Data Handling, Ethical Considerations.
- Prioritization: Start simple
→ scale up with advanced models → deploy → add fairness checks.

Academic Project (2024) Update: Credit Card Approval Prediction



I completed an academic project on predicting credit card approvals using **Machine Learning**, based on data like income, employment, credit history, and debt-to-income ratio.



I explored **Random Forest**, **Logistic Regression**, and **Decision Trees** to find the most accurate model—helping banks make faster, smarter, and safer decisions.

#MachineLearning #Python #DataScience
#CreditRisk #RandomForest #AcademicProject

Data Preprocessing

Matter:

- Handle missing values (income, age, credit history).

- Encode categorical variables (gender, job type).
- Scale numerical features (income, loan amount).
- Split dataset into train/test sets (80/20).



Model Development

Matter:

- Train Logistic Regression as baseline.
- Compare Random Forest and XGBoost for accuracy.
- Evaluate with metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC.
- Select best-performing model for deployment.

Deployment (Flask + Frontend)

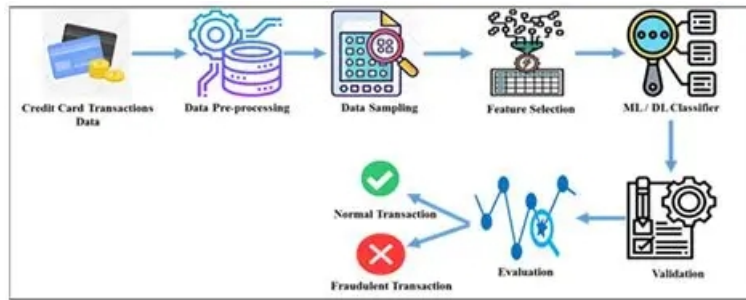
Matter:

- Flask API with /predict endpoint.
- Web form (HTML/CSS/JS) for user input.
- Integration: User submits data → API returns Approved/Not Approved.
- Host locally or on cloud (Heroku/Render).

Testing

Matter:

- Unit tests for preprocessing functions.
- Integration tests for frontend → backend → model pipeline.
- User testing with different applicant profiles.
- Validate fairness across demographic groups.



Documentation

Matter:

- README explaining dataset, preprocessing, model choice, and deployment steps.
- User manual for running the system.
- Technical documentation for developers.
- Screenshots of workflow and outputs.

Demonstration

Matter:

- Final presentation with slides showing workflow, model performance, and demo screenshots.
- Live demo: User enters applicant details → system predicts approval.
- Showcase fairness metrics and accuracy scores.